STATISTICAL ANALYSIS OF

INDIA POPULATION DATA

Census of India 2011 — Data Science Research Report

Submitted by: Amit Kumar

BTech CSE Core | Data Science

Abstract

India's 2011 Census is, frankly, one of the most ambitious data collection exercises ever attempted by any country. Over 2.7 million enumerators fanned out across 640,000 villages and nearly 8,000 towns to count every person in a country of 1.21 billion. What came back was a 98-column dataset covering everything from household size and literacy to caste categories and employment type. This report works through that data using descriptive statistics, comparative state analysis, and visualisations to answer a few simple but important questions: how is India's population distributed, who's literate, who's working, and where are the biggest inequalities hiding?

The short version of what the data says: India made real progress between 2001 and 2011 — literacy jumped from 52.21% to 63.07%, urbanisation increased, and more people moved into non-agricultural employment. But the gains were deeply uneven. Bihar's literacy sat at 50.44% while Kerala's was 84.22%. The child sex ratio dropped for the third census in a row, down to 918 from 945 in 1991. Female worker participation was 25.51% against male participation of 53.26%. And over 305 million people from Scheduled Caste and Scheduled Tribe communities were still visibly lagging every major indicator despite decades of reservation policy.

Seven original charts and several data tables are used throughout to make these patterns visible rather than just listing numbers. The report closes with what the data collectively argues for — which, to put it bluntly, is more focused, spatially targeted policy rather than the same national schemes applied everywhere regardless of how different the starting conditions are.

Keywords: Census of India 2011, demographic analysis, literacy rate, sex ratio, child sex ratio, worker participation, SC/ST population, EAG states, rural-urban divide, gender equity.

1. Introduction

1.1 What the Census Actually Is

The Census of India happens once every decade under the Census Act, 1948. The 2011 edition was the 15th in a series going back to 1872, and the 7th since independence. It's not just a headcount — it asks about household amenities, employment type, caste category, disability, and education level. That depth is why it remains the most reliable demographic source available for planning purposes.

The data it produces feeds directly into Finance Commission allocations (how much money each state gets from the centre), constituency delimitation (how many parliamentary seats a state has), and the design of schemes like the Mid-Day Meal programme, MGNREGS, and National Health Mission. Get the data wrong — or fail to analyse it carefully — and the planning built on top of it goes wrong too.

1.2 Dataset Details

Attribute	Value	Details
Source	Census of India 2011	Office of RGI, Ministry of Home Affairs
Total Records	108 rows	Each state/UT split into Total, Rural, Urban
Variables	98 columns	Population, literacy, employment, SC/ST, age groups
Geographic Scope	35 States & UTs	Full national coverage — all administrative units
Census Year	2011	15th decennial census; 7th post-independence
Field Workforce	2.7 million enumerators	Largest administrative exercise in the world

1.3 Research Objectives

1. Map India's population distribution and understand which states are statistical outliers
2. Measure literacy, sex ratio, and worker participation gaps between states and between genders
3. Analyse the rural-urban split and its implications for service delivery
4. Track child sex ratio trends across three censuses (1991, 2001, 2011)
5. Assess the demographic situation of SC and ST communities
6. Derive specific, data-backed policy priorities from the findings

2. Methodology

The raw Primary Census Abstract data (released in .xls format by the Office of the Registrar General) was cleaned and structured in Python using Pandas. Rows were filtered by the 'Level' and 'TRU' columns to isolate what was needed at each stage of the analysis — national totals, state-level figures, and rural/urban breakdowns. No significant missing values appeared in the primary demographic columns. States like Punjab and Chandigarh show zero ST population, which accurately reflects the absence of notified Scheduled Tribes in those regions — not a data gap.

All rates and ratios were computed from scratch rather than taken directly from census summaries, to ensure consistency across metrics. The key derived measures are:

Measure	Formula	What It Captures
Sex Ratio	(Female Pop ÷ Male Pop) × 1000	Females per 1,000 males — gender balance indicator
Child Sex Ratio	(Girls 0–6 ÷ Boys 0–6) × 1000	Gender balance at birth — detects sex selection
Literacy Rate	(Literate Pop ÷ Pop aged 7+) × 100	% of 7+ population who can read & write
Worker Part. Rate	(Total Workers ÷ Total Pop) × 100	Share of population in paid/productive work
Standard Deviation	$\sqrt{[\Sigma(x_i - \bar{x})^2 \div N]}$	Spread of values — measures inter-state inequality
Arithmetic Mean	$\Sigma(x_i) \div n$	Central tendency used for national benchmarking

All seven charts in this report use consistent colour coding: green for above-average performance, red/orange for below-average. Reference lines on each chart mark the national average so state-level comparison is immediate.

3. Results and Analysis

3.1 National Snapshot

121.05 Cr	63.07%	943	918	39.79%
Total Population	**Literacy Rate**	**Sex Ratio**	**Child Sex Ratio**	**Worker Part. Rate**
2nd largest in world, 2011	*Up 10.86 pp from 2001*	*Below bio. norm of 950–980*	*3rd consecutive census decline*	*Female WPR: only 25.51%*

A few numbers from this table deserve more than a glance. The child sex ratio of 918 isn't just below 943 — it's the third time in a row it's dropped across censuses, and that trend line is more alarming than any single snapshot. The literacy figure of 63.07% represents nearly 11 points of progress in 10 years, which sounds impressive until you do the maths: it still leaves 447 million people unable to read or write. And the WPR of 39.79% looks reasonable until you split it by gender — male 53.26%, female 25.51% — a gap of nearly 28 percentage points that makes the national figure fairly misleading on its own.

Indicator	Value	2001 Figure	Change
Total Population	1,21,05,69,573	1,02,87,37,436	+17.64%
Rural Population	83,34,63,448 (68.85%)	74.27% in 2001	Declining
Sex Ratio	943	933	+10 ↑
Child Sex Ratio (0–6 yrs)	918	927	−9 ↓
Literacy Rate	63.07%	52.21%	+10.86 pp ↑
SC Population Share	16.63% (20.13 Cr)	16.20%	Stable
ST Population Share	8.61% (10.43 Cr)	8.20%	Slight rise
Total Workers	48.17 Cr (39.79%)	40.23 Cr (39.26%)	+0.53 pp

3.2 Population Distribution — Extreme Inequality

The mean state population is 3.46 crore. The standard deviation is 4.77 crore — larger than the mean itself. That's a statistical way of saying the data isn't just skewed, it's dominated by a handful of enormous outliers. Uttar Pradesh at 19.98 crore is 309 times larger than Lakshadweep at 64,473. The top four states — UP, Maharashtra, Bihar, West Bengal — hold about 35% of India's entire population.

The eight EAG states (UP, Bihar, Rajasthan, MP, Odisha, Jharkhand, Chhattisgarh, Uttarakhand) together hold over 45% of India's population while consistently scoring below the national average on literacy, sex ratio, and health outcomes. The 'BIMARU' label for Bihar, MP, Rajasthan, and UP is sometimes dismissed as old stereotyping, but the Census 2011 data gives it statistical backing.

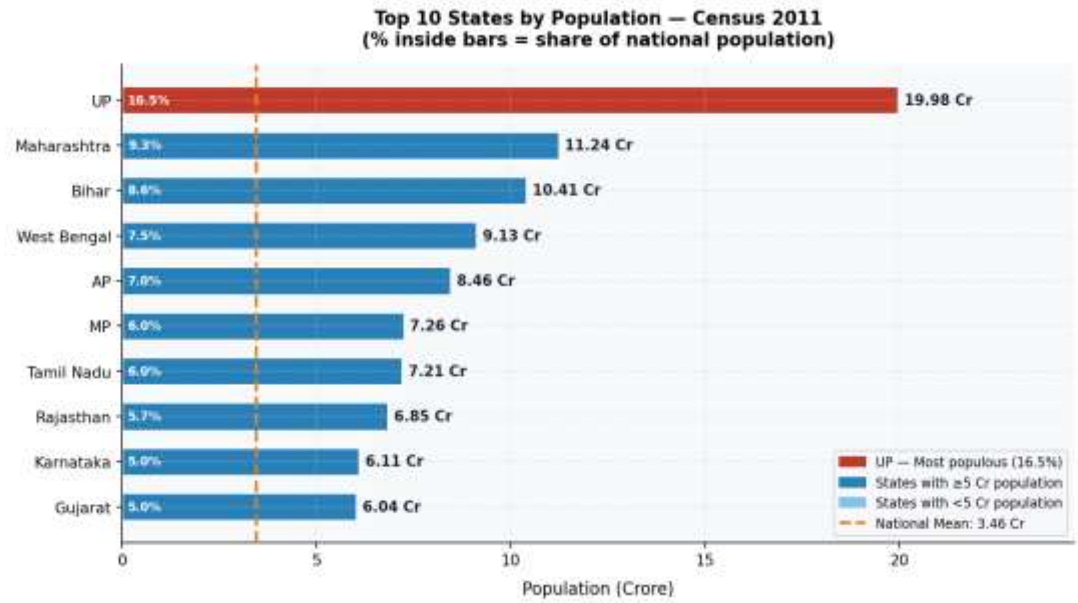

Figure 1 — Population: Top 10 States. Percentages inside bars = share of national population. Dashed line = national mean (3.46 Cr).

3.3 Literacy — Strong Progress, Deep Inequality

Literacy is probably the indicator most directly tied to long-term development outcomes — it affects health, income, civic participation, and fertility rates all at once. The national rate of 63.07% in 2011 was a meaningful improvement from 52.21% in 2001, driven partly by the Sarva Shiksha Abhiyan and expanded school infrastructure. But the improvement wasn't uniform. States that were already ahead pulled further ahead; states that were behind didn't close the gap proportionately.

The range is striking: Kerala at 84.22%, Bihar at 50.44%. A 33.78 percentage-point spread between Indian states on something as fundamental as reading and writing is hard to overstate. The chart below covers 22 states and union territories — look at how many of them cluster below the 63% line, and how few are near the 75% target that development planners consider meaningful functional literacy.

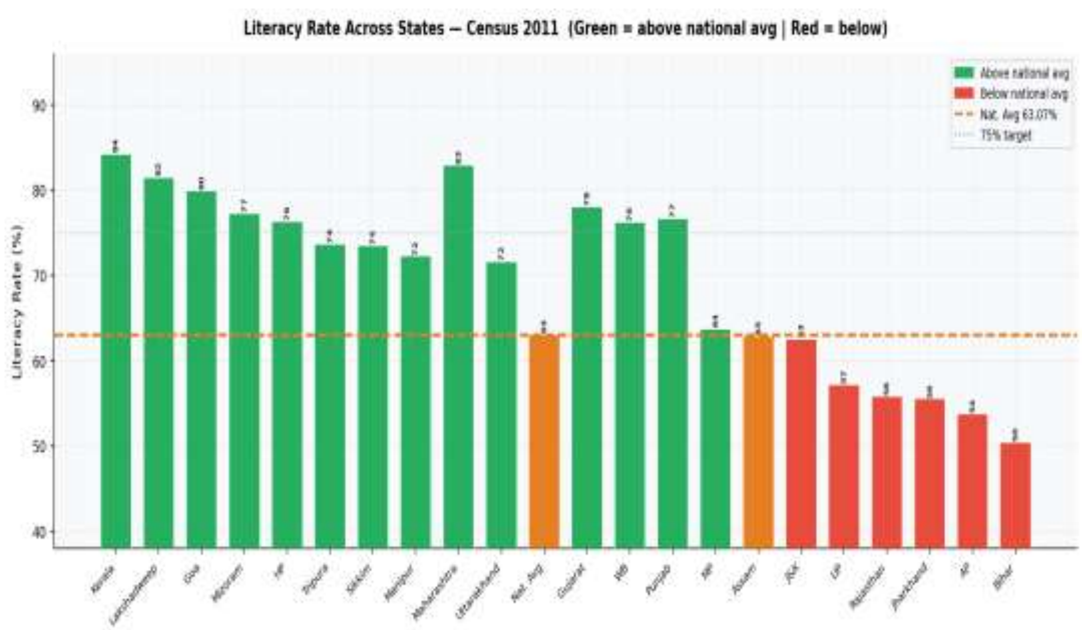

Figure 2 — Literacy rates across 22 states/UTs. Green = above national avg (63.07%). Red = below. Dotted line = 75% target.

3.4 Gender Literacy Gap — 16 Points That Matter

The national gender gap in literacy — 80.89% for men, 64.64% for women — is 16.25 percentage points. That gap is the accumulated result of decades of under-investment in girls' education: fewer schools within walking distance in rural areas, safety concerns about girls commuting, early marriage pulling girls out before they complete schooling, and households prioritising sons' education when budgets are tight.

Rajasthan's gender gap exceeds 27 points — the worst among large states. Kerala's is under 4 points — the best in the country, and the result of public schooling that dates back to the Travancore and Cochin kingdoms in the 19th century. The lower panel in the chart below shows the actual gap for each state colour-coded by severity. It's worth noting that several states with relatively high male literacy still have large gaps — meaning male literacy alone doesn't close gender inequality.

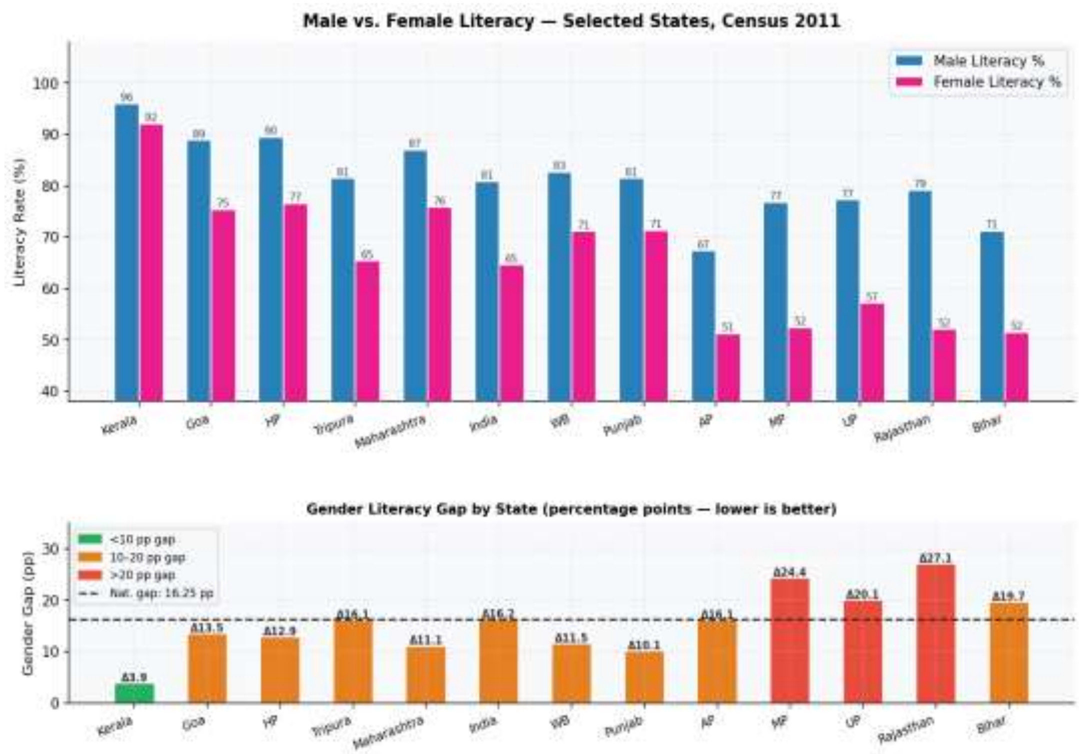

Figure 3 — Top panel: Male vs. Female literacy by state. Bottom panel: Gender gap in percentage points (Δ). Green < 10 pp, Orange 10–20 pp, Red > 20 pp.

3.5 Sex Ratio — and the Alarming Child Sex Ratio Trend

India's national sex ratio of 943 females per 1,000 males is below the biological expectation of 950–980. The adult ratio has actually been improving slightly — it was 933 in 2001 — but the child sex ratio has been moving in the opposite direction, which is the more important number.

The child sex ratio (0–6 age group) in 2011 was 918 — down from 927 in 2001 and 945 in 1991. **That is three consecutive census periods of decline.** If you're looking for one number in this entire dataset that signals a serious structural problem, this is probably it.

Haryana is the most extreme case: child sex ratio of 830, meaning only 830 girls for every 1,000 boys born/surviving. And Haryana isn't poor — its literacy rate is 75.55% and per-capita income is above the national average. This is a culture and law-enforcement failure, not just a poverty outcome. The PC-PNDT Act has been law since 1994 and clearly hasn't been enforced consistently enough. Chart 4 compares both adult and child sex ratios side by side — the gap between them in states like Haryana and Punjab is particularly visible.

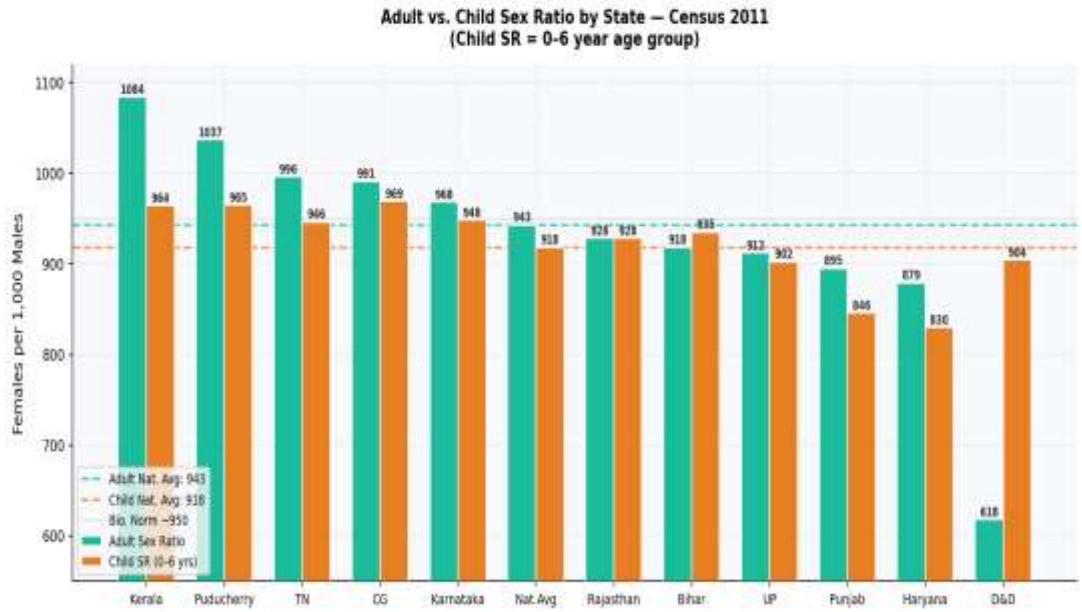

Figure 4 — Adult sex ratio vs. Child sex ratio (0–6 years) by state. Dashed lines = respective national averages.

3.6 Child Sex Ratio Across Three Censuses (1991–2011)

This is worth its own chart. Tracking the child sex ratio across 1991, 2001, and 2011 shows which states have been improving, which have been declining, and which have been on a steady downward slide. Nationally, the trend is clearly negative. Most worrying is that the decline is visible even in states that aren't particularly poor — Punjab, Haryana, Gujarat — which makes poverty reduction alone an insufficient response.

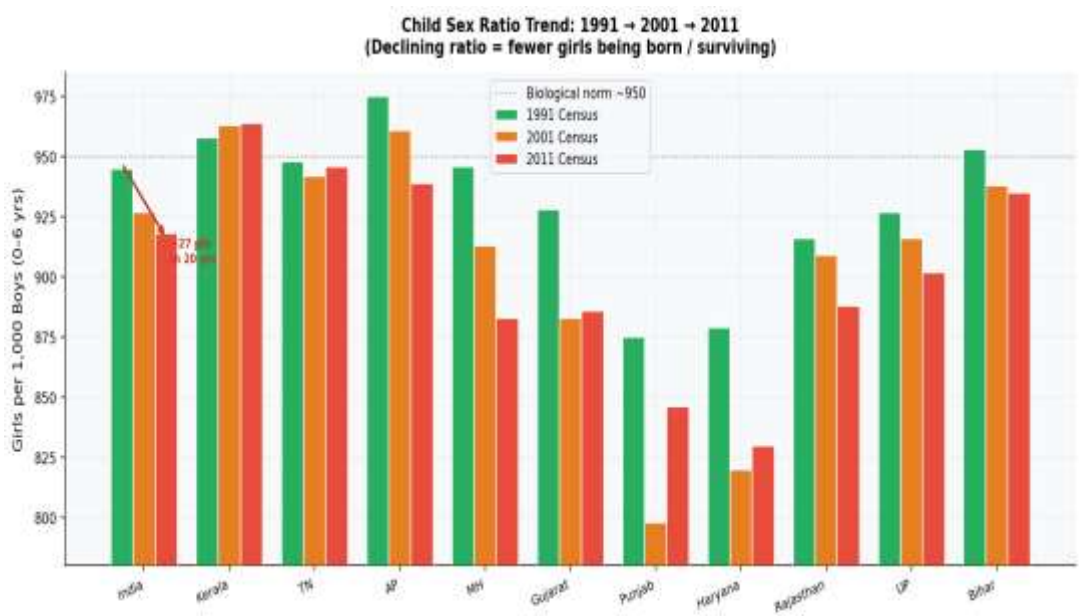

Figure 5 — Child Sex Ratio trend: 1991 → 2001 → 2011. Arrow shows India's 27-point national decline over 20 years.

The arrow on India's national bar in this chart covers a 27-point drop over 20 years. Some states — Kerala, Tamil Nadu, Chhattisgarh — have held relatively stable or even improved slightly. Others like Haryana (879→820→830) show a decline that levelled off only marginally. Bihar and UP remain concerningly low despite being states where absolute poverty is higher, suggesting that son-preference cuts across economic lines.

3.7 Employment Structure — Agriculture Still Dominates

Of India's 1.21 billion people, 481.7 million (39.79%) are classified as workers. Agriculture remains the single largest employment category — cultivators and agricultural labourers together account for 50.7% of all main workers. The 'Other Workers' category (trade, services, construction, government) has grown to 34.9% of main workers from around 22% in 2001, which is the most direct reflection of India's expanding urban economy.

One figure that often goes unremarked: 24.77% of all workers — roughly 119 million people — are marginal workers who worked fewer than 183 days in the reference year. These are the seasonal farm hands, construction daily-wage workers, and informal street vendors who have no income security, no employment benefits, and no safety net beyond MGNREGS. That's over 100 million people in economically precarious employment.

Worker Category	Count	% of Pop.	% of Workers
Main Workers	**36,24,46,420**	**29.94%**	**75.23%**
— Cultivators	9,58,41,357	7.92%	19.90%
— Agricultural Labourers	8,61,66,871	7.12%	17.89%
— Household Industry Workers	1,23,31,464	1.02%	2.56%
— Other Workers (Services/Trade)	16,81,06,728	13.89%	34.90%
Marginal Workers	**11,92,96,891**	**9.85%**	**24.77%**
Non-Workers	**72,88,26,262**	**60.21%**	**—**

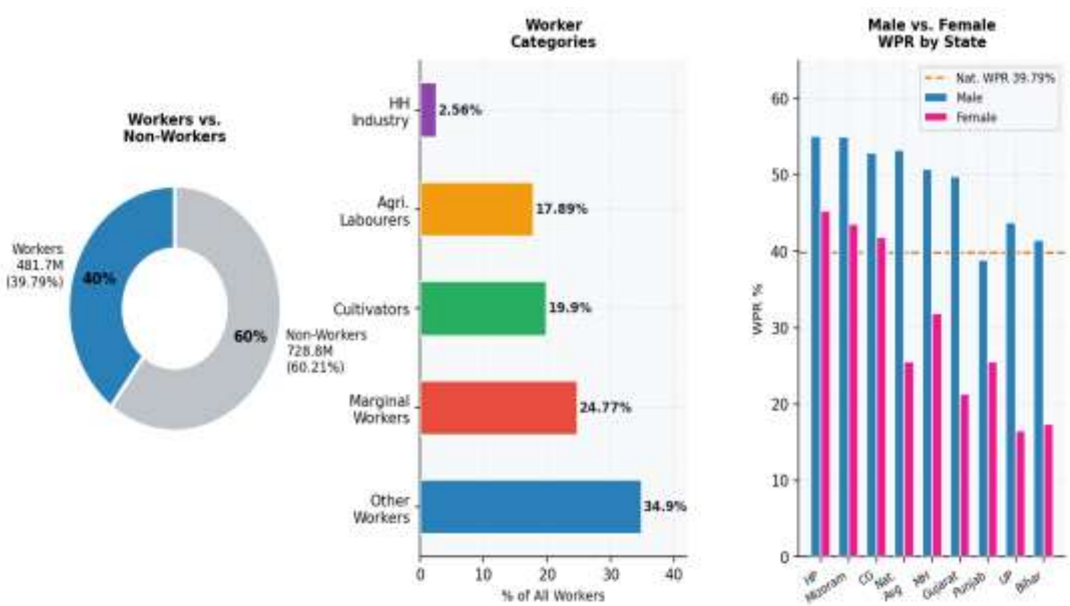

Figure 6 — Left: Worker/Non-worker split. Centre: Worker category breakdown by % share. Right: Male vs. Female WPR across states.

The right panel of Figure 6 shows the male-female WPR gap by state. Even Himachal Pradesh — the best performer on female WPR at around 45% — still has a visible gap below male WPR. In UP and Bihar, female WPR barely clears 16–17%. The national figures (male 53.26%, female 25.51%) are a 28-point gap that represents a massive amount of women's economic potential that goes uncounted — because much of what women actually do, from unpaid agricultural support to household labour, falls outside the census definition of 'worker'.

3.8 Rural-Urban Divide and SC/ST Distribution

India was 68.85% rural in 2011. That single number shapes almost every delivery challenge the government faces — schools, hospitals, banking, broadband, all of it is harder to reach when 833 million people are spread across 640,000 villages. The variation is extreme: Delhi is 97.5% urban, Himachal Pradesh is 90% rural, Bihar is 88.7% rural. Figure 7 (left panel) shows this spread.

The right panel plots SC and ST population shares side-by-side for selected states. The contrast is stark: Punjab has the highest SC concentration (31.9%) and effectively zero ST population. Mizoram is the reverse — 94.4% Scheduled Tribe, minimal SC. These patterns aren't incidental — they reflect centuries of migration, settlement history, and who got displaced from which land. Understanding this geography is essential for targeting welfare programmes correctly.

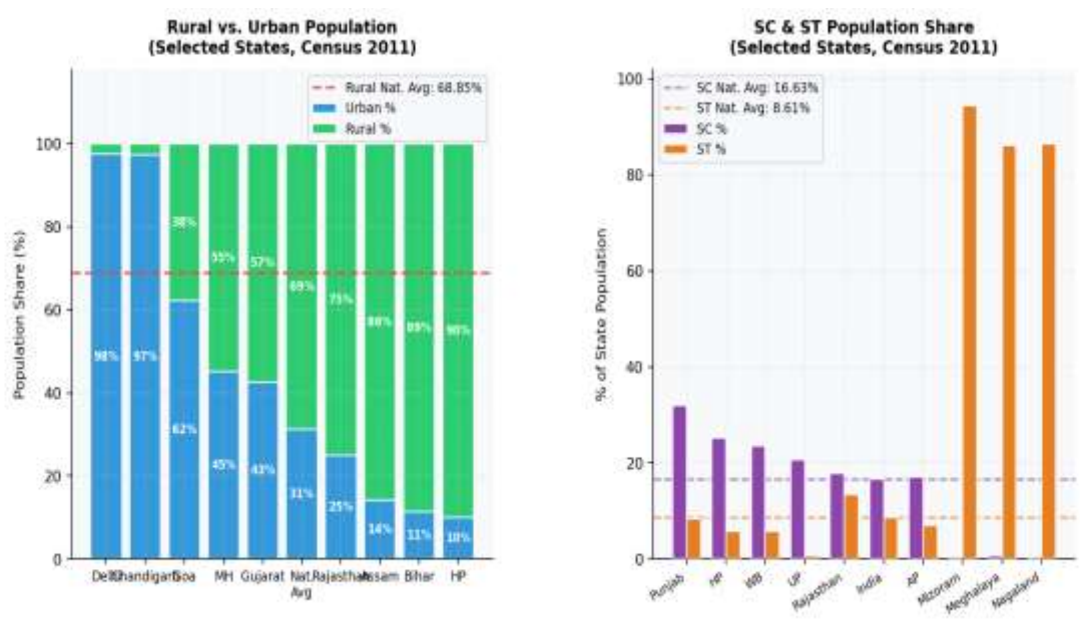

Figure 7 — Left: Rural vs. Urban split by state (% inside bars). Right: SC and ST population share, selected states.

Together, SC (16.63%, 201 million) and ST (8.61%, 104 million) populations make up over 305 million people — more than the entire population of the United States. Despite decades of reservation in government employment and higher education, both groups still trail national averages on literacy, asset ownership, and health outcomes. The data doesn't capture quality-of-life directly, but the pattern is consistent enough across multiple indicators to be statistically meaningful.

4. Comparative State Scorecards

The table below puts the key indicators side by side across selected states. Reading it as a whole reveals two things: the North-South development divide is real and consistent, and economic size doesn't predict demographic outcomes — Himachal Pradesh (0.69 crore people) performs better on sex ratio and female WPR than Maharashtra (11.24 crore).

State	Pop (Cr)	Literacy %	Sex Ratio	CSR (0–6)	WPR %	Overall
Kerala	3.34	84.22	1,084	964	38.5	**Excellent**
Tamil Nadu	7.21	80.33	996	946	40.8	Very Good
Maharashtra	11.24	82.91	929	883	42.1	Good
Himachal P.	0.69	76.34	972	909	51.9	Good
Gujarat	6.04	78.03	919	886	43.8	Good
Punjab	2.77	76.68	895	846	38.9	Mixed
National Avg	**—**	**63.07**	**943**	**918**	**39.8**	**Benchmark**
West Bengal	9.13	76.26	950	956	38.5	Average
Rajasthan	6.85	66.11	928	888	42.1	Below Avg
Uttar Pradesh	19.98	67.73	912	902	33.8	Below Avg
Haryana	2.54	75.55	879	830	40.2	Poor CSR
Bihar	10.41	61.80	918	935	34.7	**Poor**

Punjab is the most interesting anomaly in this table. Its literacy rate is 76.68% — well above the national average. Its sex ratio is 895 — well below. And its child sex ratio is 846 — one of the worst in the country. That disconnect between education and gender equity shows something important: literacy by itself doesn't change gender norms. Punjab's land inheritance traditions and dowry practices haven't been touched by rising male or female literacy, because the economic incentives driving son-preference are still in place.

5. Discussion and Policy Implications

5.1 The EAG State Problem Is Structural

The eight EAG states — UP, Bihar, Rajasthan, MP, Odisha, Jharkhand, Chhattisgarh, Uttarakhand — together hold over 45% of India's population and consistently underperform across every indicator in this dataset. This isn't just a resource allocation problem. These states receive large central transfers but have historically struggled to convert them into ground-level outcomes, partly because of governance capacity gaps, partly because social norms around caste and gender are more entrenched, and partly because baseline infrastructure — roads, schools, health centres — was so weak to begin with.

A national scheme designed around Kerala's conditions will predictably underperform in Bihar. The scale of this problem — involving over 500 million people — means it has to be treated as a distinct policy domain, not a regional variation of the national average.

5.2 Three Non-Negotiable Gender Priorities

- PC-PNDT Act enforcement: The child sex ratio has declined across three censuses. The law banning sex determination tests for sex selection exists. The problem is enforcement. States like Haryana need district-by-district enforcement drives backed by real consequences for violators — not just awareness campaigns.
- Girls' school completion: Rajasthan's 27-point gender literacy gap doesn't close by building more schools alone. Conditional cash transfers tied to secondary school completion (not just enrolment), safe hostel accommodation in rural areas, and cycle-to-school programmes have all shown measurable impact where properly implemented.
- Female WPR: A female WPR of 25.51% represents an enormous pool of productive capacity going untapped. Safe transport, childcare at worksites, and skill training for non-agricultural employment in rural areas are the interventions with the strongest evidence base.

5.3 Marginal Workers Need Direct Attention

119 million marginal workers — working fewer than 183 days a year — are in a policy gap. MGNREGS reaches some of them, but its rural focus and dependence on state implementation means coverage is patchy. Extending portable social security (health insurance, provident fund contributions) to informal workers, regardless of whether their employer is registered, is the kind of structural change that would actually improve their situation.

6. Conclusion

Census 2011 gives you a detailed photograph of India at a specific moment — and what it shows is a country at a genuine inflection point. The progress is real: literacy up 11 points in a decade, more people in non-agricultural work, urban areas expanding, the adult sex ratio nudging upward. These aren't trivial achievements in a country of 1.21 billion with the kind of internal diversity India has.

But the photograph also shows the cracks clearly. Bihar's 50.44% literacy and Kerala's 84.22% aren't two data points in the same distribution — they're descriptions of two completely different states of human development coexisting within a single federal system. Haryana's child sex ratio of 830 in a state with 75% literacy tells you that education and gender equity are not the same thing, and that assuming economic progress will fix social discrimination on its own is a bet the data doesn't support.

The three findings from this analysis that carry the most policy weight are: first, the declining child sex ratio — three censuses of decline with no reversal — which is a trend with serious long-run demographic consequences; second, the 28-point gender WPR gap, which represents one of the largest untapped sources of economic potential in any country; and third, the concentration of developmental deficit in EAG states holding nearly half the country's population, which makes spatially targeted policy not just preferable but necessary.

The national averages that get cited in policy documents — 63% literacy, 943 sex ratio, 39.79% WPR — are mathematically accurate but strategically misleading if they're used to design uniform national responses. The real story is in the distribution: who is below the average, by how much, and why. That's what this analysis has tried to surface, and it's what any serious development planning exercise should be doing with this data.

Census 2021, when its full dataset is released, will show whether any of these trends shifted in the decade that included demonetisation, GST rollout, COVID-19, and a series of targeted gender programmes. A time-series comparison across 2001, 2011, and 2021 would be the logical — and arguably urgent — next step.

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